Weikai Lin

CONTACT INFORMATION

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University of Rochester
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https://linwk20.github.io/
Google Scholar | LinkedIn

RESEARCH INTERESTS

I am interested in end-to-end hardware–software co-design. My publications span AR/VR, AI, and imaging systems. I am also an AI maximalist interested in Autoresearch and AI4S, aiming to accelerate scientific discovery by orders of magnitude and unlock research that was previously impossible.

- **AR/VR systems:** system-level hardware–software co-design for efficient immersive systems. [SIGGRAPH Asia'26, TVCG'26, CVPR'26, 2×arXiv'26, SIGGRAPH Asia'25, 2×ASPLOS'25, ISCA'25, HotMobile'25, TACO'24, arXiv'24]
- **AI systems:** generative, multimodal, and embodied AI. [ECCV'26, 2×NeurIPS'26, ACM MM'26, EMNLP'26, 2×arXiv'26, IROS'24, RO-MAN'23, PKU ASN'23, 2×ICDL'21]
- **Imaging systems:** co-design of sensors and optics with downstream vision models. [2×DAC'26, DAC'25, WACV'25]

EDUCATION

University of Rochester, Rochester, NY, USA
Ph.D., Computer Science — Aug 2023 – May 2028 (expected)
M.Sc., Computer Science — Aug 2023 – May 2025
Advisor: Yuhao Zhu

Peking University, Beijing, China
M.Sc., Intelligence Science and Technology — Sep 2020 – Jun 2023
Advisor: Dingsheng Luo

Tsinghua University, Beijing, China
B.E., Electronic Engineering — Aug 2016 – Jun 2020
Advisor: Li Su

HONORS AND AWARDS

- **2026, ACM/IEEE DAC Young Fellow**
- **2025, ACM ASPLOS Best Paper Award**
  "MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering" (first author)
- **2025, OpenAI Research Grant ($5K)**

PROFESSIONAL POSITIONS

Meta Platforms, Inc., CA, USA
Research Scientist Intern — May 2026 – Dec 2026
Topics: Efficient large-scale view synthesis models [Project page].

Pixocial AI Lab, Bellevue, WA, USA
Applied Research Intern — May 2025 – Aug 2025
Mentor: Haoxiang Li.
Topics: Physics-based video diffusion [Project page, accepted to ECCV'26].

Advanced Micro Devices, Inc. (AMD), Beijing, China
Student Intern — Jul 2022 – Feb 2023

Mentor: Fuwei Yang.
Topics: SLAM and DNN evaluations on AMD GPUs. Participated in a ROCm release.

Cambricon Technologies Co., Ltd., Beijing, China
Student Intern Jul 2019 – Aug 2019
Mentor: Miao Li.
Topics: Neural network pruning.

PUBLICATIONS

* denotes equal contribution.

- Weikai Lin, Nicholas Milef, Carl S. Marshall, Grant Kaijuin Yang, Todd Keeler, Yuhao Zhu
  "GLoD-LVSM: Learning Geometry-Grounded Continuous-LoD Scene Representations for Large View Synthesis Models"
  [arXiv 2026] *arXiv:2610.20254*

- Weikai Lin, Junjie Zhao, Carl Marshall, Sushant Kondguli, Yuhao Zhu
  "ControlGS: Conditioning Neural Gaussians for Downstream-Processing–Aware XR Rendering"
  [SIGGRAPH Asia 2026] *SIGGRAPH Asia Conference Papers*

- Weikai Lin, Yu Feng
  "MetaSapiens v2: Advancing Real-Time Foveated Neural Rendering via Foveation-Aware Pruning and Stereo Warping"
  [arXiv 2026] *arXiv:2608.17969*

- Weikai Lin, Sheng Zhao, Ian Ross, Carl Marshall, Sushant Kondguli, Yuhao Zhu
  "LowPowAR: Power-Constrained Tone Mapping for Augmented Reality"
  [TVCG 2026] *IEEE Transactions on Visualization and Computer Graphics* (ISMAR Journal Track)

- Weikai Lin, Haoxiang Li, Yuhao Zhu
  "ControlHair: Synergizing Physics Simulator and Video Diffusion for Controllable Dynamic Hair Rendering"
  [ECCV 2026] *19th European Conference on Computer Vision*

- Sheng Zhao, Weikai Lin, Yuhao Zhu
  "GazeFlow: From Human Gaze Behavior to Generative Egocentric Gaze Prediction"
  [NeurIPS 2026] *The Fortieth Annual Conference on Neural Information Processing Systems*

- Sheng Zhao, Weikai Lin, Yuhao Zhu
  "Multidimensional Observer Model and Perceptual Dimensions of Human Image Quality Assessment"
  [NeurIPS 2026] *The Fortieth Annual Conference on Neural Information Processing Systems*

- Zijian Gu, Weikai Lin, Shuang Zhou, Zihan Chen, Song Wang
  "UniMod: Enhancing Multi-Modal Medical Diagnosis through Cross-Modality and Within-Modality Alignment"
  [ACM MM 2026] *34th ACM International Conference on Multimedia*

- Ruofan Xing, Arman Akbari, Weikai Lin, Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
  "HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training"
  [DAC 2026] *63rd ACM/IEEE Design Automation Conference*

- Han Yan*, Yifei Zou*, Weikai Lin, Yuhao Zhu, Xuan Zhang
  "Invited Paper: A Quantitative Approach to Exploring Novel Image Sensor Architecture Towards Autonomous Edge Machine Vision"
  [DAC 2026] *63rd ACM/IEEE Design Automation Conference*

- He Zhu*, Xiaotong Huang*, Zihan Liu, Weikai Lin, Xiaohong Liu, Zhezhi He, Jingwen

Leng, Minyi Guo, Yu Feng
"SeeLe: A Unified Acceleration Framework for Real-Time Gaussian Splatting on Mobile Devices"
[CVPR 2026 Highlight] *IEEE/CVF Conference on Computer Vision and Pattern Recognition*

- Tianyu Hu, Weikai Lin, Weizhi Zhang, Jing Ma, Song Wang
"MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents"
[arXiv 2026] *arXiv:2605.00356*

- Bo Yan, Weikai Lin, Yada Zhu, Song Wang
"SafeDream: Safety World Model for Proactive Early Jailbreak Detection"
[arXiv 2026] *arXiv:2604.16824*

- Bo Yan, Weikai Lin, Song Wang
"The Menu Is an Execution Prior: State-Path Tool Menus for Online Agents"
[EMNLP 2026 Main] *Conference on Empirical Methods in Natural Language Processing*

- Weikai Lin, Sushant Kondguli, Carll Marshall, Yuhao Zhu
"PowerGS: Display–Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems"
[SIGGRAPH Asia 2025] *SIGGRAPH Asia Conference Papers*

- Weikai Lin*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
"SnapPix: Efficient-Coding–Inspired In-Sensor Compression for Edge Vision"
[DAC 2025] *62nd ACM/IEEE Design Automation Conference*

- Weikai Lin*, Yu Feng*, Yuhao Zhu
"MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering"
[ASPLOS 2025 Best Paper Award] *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*

- Yu Feng*, Zheng Liu*, Weikai Lin*, Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He, Jieru Zhao, Yuhao Zhu
"StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination"
[ASPLOS 2025] *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*

- Yu Feng, Weikai Lin, Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, Yuhao Zhu
"Lumina: Real-Time Mobile Neural Rendering by Exploiting Computational Redundancy"
[ISCA 2025] *52nd IEEE/ACM International Symposium on Computer Architecture*

- Adith Boloor, Weikai Lin, Tianrui Ma, Yu Feng, Yuhao Zhu, Xuan Zhang
"Private-Eye: In-Sensor Privacy Preservation Through Optical Feature Separation"
[WACV 2025] *IEEE/CVF Winter Conference on Applications of Computer Vision*

- Nan Wu*, Weikai Lin*, Ruizhi Cheng*, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
"Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting"
[HotMobile 2025] *26th International Workshop on Mobile Computing Systems and Applications*

- Yu Feng*, Weikai Lin*, Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, Yuhao Zhu
"Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture"
[TACO 2024] *ACM Transactions on Architecture and Code Optimization*

- Wenfei Hu, Weikai Lin, Hongyu Fang, Yi Wang, Dingsheng Luo

"OW3Det: Toward Open-World 3D Object Detection for Autonomous Driving"
[IROS 2024] *IEEE/RSJ International Conference on Intelligent Robots and Systems*

- Weikai Lin, Yu Feng, Yuhao Zhu
"RTGS: Enabling Real-Time Gaussian Splatting on Mobile Devices Using Efficiency-Guided Pruning and Foveated Rendering"
[arXiv 2024] *arXiv:2407.00435* (preprint and shortened version of MetaSapiens)

- Wenfei Hu, Weikai Lin, Hongyu Fang, Yi Wang, Dingsheng Luo
"Learning Clear Class Separation for Open-Set 3D Detector in Autonomous Vehicle Via Selective Forgetting"
[RO-MAN 2023] *32nd IEEE International Conference on Robot and Human Interactive Communication*

- Weiming Qu, Tianlin Liu, Weikai Lin, Dingsheng Luo
"A Review of Robot Learning"
[Acta 2023] *Acta Scientiarum Naturalium Universitatis Pekinensis*, 59(6):1069–1086. DOI: 10.13209/j.0479-8023.2023.086

- Shuai Fang, Yaoyao Wei, Weikai Lin, Jianan Zhang, Tianlin Liu, Dingsheng Luo
"Approaching Sound Object with Sensorimotor Coordination when Sensors Partially Damaged"
[ICDL 2021] *IEEE International Conference on Development and Learning*

- Yaoyao Wei, Shuai Fang, Weikai Lin, Jianan Zhang, Dingsheng Luo
"Acquiring Robot Navigation Skill with Knowledge Learned from Demonstration"
[ICDL 2021] *IEEE International Conference on Development and Learning*

WORKSHOPS

- Weikai Lin*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
"SnapPix: Efficient-Coding–Inspired In-Sensor Compression for Edge Vision"
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [link]

- Ruofan Xing, Arman Akbari, Weikai Lin, Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
"HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training"
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [link]

- Han Yan, Yifei Zou, Weikai Lin, Xuan Zhang, Yuhao Zhu
"Do Energy-Harvesting Pixels Reduce Carbon Cost of Visual Computing Systems?"
[HotEthics 2026] *Workshop on Ethical Systems and Architecture Design 2026* [link]

PATENTS AND OTHER PROJECTS

- Dingsheng Luo, Xihong Wu, Yifan Yuan, Wenfei Hu, Weiming Qu, Yudi Zou, Jiawen Wang, Hongyu Fang, Weikai Lin
Control Method and System for Improving Operation Precision of Robot Arm
[Patent] *CN202310068025.4*, China; published as CN116079730A, 2023

- Dingsheng Luo, Xihong Wu, Shuai Fang, Jianan Zhang, Weikai Lin, Tianlin Liu
Active Auditory Positioning Method for Map-Free Navigation
[Patent] *CN202210079214.7*, China; published as CN114563011A, 2022

- Dingsheng Luo, Xihong Wu, Jianan Zhang, Shuai Fang, Tianlin Liu, Weikai Lin, Hongyu Fang
Robot Map-Free Navigation Method Based on Time Sequence Information Modeling
[Patent] *CN202110018866.5*, China; published as CN112857370A, 2021

- "Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality: An FPGA Demo"
[Open-Source Project] Open-sourced project with optimized CPU/GPU/FPGA implementations. [code]

TEACHING EXPERIENCE

Teaching Assistant, University of Rochester

- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 257/457 *Computer Networks*

EXTERNAL SERVICE

Reviewer

- SIGGRAPH: 2025, 2026
- SIGGRAPH Asia: 2025
- NeurIPS: 2026
- ICLR: 2027
- IEEE VR: 2027
- ISMAR: 2026
- WACV: 2026, 2027
- TCAD: 2025
- JCST: 2026